1 Problem Statement

Let $f : \mathbb{S}^2 \to \mathbb{S}^2$ be a branched cover, $V(f)$ be the set of its critical values, and $A, B \subset \mathbb{S}^2$ be finite sets.

1. If $f(A) \cup V(f) \subset B$, then every complex structure on $(\mathbb{S}^2, B)$ can be pulled back to a complex structure on $(\mathbb{S}^2, A)$, and this induces the map $f^* : \mathrm{Teich}(B) \to \mathrm{Teich}(A)$.

2. If $A \subset B$, then forgetting the points $B \backslash A$ induces the map $i_* : \mathrm{Teich}(B) \to \mathrm{Teich}(A)$.

If $f(A) \cup V(f) \cup A \subset B$, then the deformation space is defined by

$$\mathcal{D}_f := \{\tau \in \mathrm{Teich}(B) : f^*(\tau) = i_*(\tau)\}.$$

Consider a quadratic rational map f with critical points c_1 and c_2. Assume that c_1 belongs to a cycle of length 4, and the critical value $v = f(c_2)$ does not belong to the cycle. By taking

$$A = \{c_1, f(c_1), f^{\circ 2}(c_1), f^{\circ 3}(c_1)\}, \quad B = A \cup \{v\},$$

we get a deformation space $\mathcal{D}_f \subset \mathrm{Teich}(B)$. Its image in the moduli space is the set of conjugacy classes of quadratic rational maps with the properties described above.

Explicitly describe a simple closed curve $\gamma \subset (\mathbb{S}^2, B)$ such that a power of the Dehn twist around this curve preserves the deformation space, and its action on the set of connected components of $\mathcal{D}_f$ has an infinite orbit.